

Database Systems and Laboratory

Airport Extra Exercise

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Extra exercise, Airport

- ▶ We want to automate a system for the management of a tourist airport.
- ▶ The system should keep track of the aeroplanes registered at the airport, their owners, the personnel on duty at the airport and the pilots (who may or may not be the owners of the aeroplanes).
- ▶ The expected requirements of the system, gathered through a series of interviews with airport personnel, are described next.

Extra exercise, Airport

- ▶ For each person, we want to store the social security number, first and last name, address, a telephone number and, optionally, an email address.
- ▶ In addition, for each airplane owner, we want to keep track of the date of the last payment of parking fees and the total amount paid up to the current date. Moreover, we also want to record the airplanes he/she currently owns as well those he/she owned in the past, with the corresponding timespans.
- ▶ For each airport employee, we want to know the salary, the aircraft models on which he is authorised to work and for which classes of operation. Note that an employee could be qualified to perform a given intervention class on a certain airplane model but not on another.
- ▶ Due to legal and insurances issues, employees can not own nor pilot aeroplanes in they airport where they work

Extra exercise, Airport

- ▶ Each intervention class is identified by a code and is characterised by a type (cleaning, maintenance, refuelling) and a description.
- ▶ In addition, we wish to keep track of the date and duration of interventions actually carried out on individual aircraft.
- ▶ Finally, consider that an intervention may be performed by one or more employees.

Extra exercise, Airport

- ▶ Of the pilots we want to store the flight licence number and hours flown on individual aircraft.
- ▶ Of each aircraft we want to store the registration number, the model and the hangar where it is normally stored. Assume that there can be several aeroplanes of the same model and that several aeroplanes can be stored in a hangar.
- ▶ The name, which uniquely identifies the aircraft, is recorded for each model along with the weight and the capacity.
- ▶ Each hangar is characterised by a number, which uniquely identifies it within the airport, a capacity (understood as the number of aircraft that can be stored within the hangar), the number of aircraft currently assigned to it and the location (x, y, z) in a 3D reference system of the airport.